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(54) Title: **Geo-medicinal intelligence system for phytochemical prediction and personalized plant recommendation**

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(57) Abstract:

ABSTRACT GEO-MEDICINAL INTELLIGENCE SYSTEM FOR PHYTOCHEMICAL PREDICTION AND PERSONALIZED PLANT RECOMMENDATION A method (100) for operating a geo-medicinal intelligence system comprises: receiving (102) user input including a geographic location, a patient profile, and a clinical condition; predicting (106) geographic potency of phytochemical compounds using a geographic potency prediction module applying Gaussian Process Regression with a radial basis function kernel and computing predicted concentrations with confidence intervals; estimating (108) personalized dosage using a personalized dosage estimation module feeding patient profile features and compound descriptors into a Bayesian Neural Network with Monte Carlo dropout sampling to compute patient-specific bioavailability; predicting (110) herb-drug interactions using an herb-drug interaction prediction module training a Graph Neural Network on a knowledge graph to compute interaction probabilities; generating (112) plant recommendations using a smart recommendation engine applying multi-objective constraint optimization across potency, safety, personalization, and availability scores; and outputting (114) a ranked decision report including recommended plants, potency scores, personalized dosages, and confidence.

FORM-2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE

SPECIFICATION

(See section 10 and rule 13)

TITLE OF INVENTION

GEO-MEDICINAL INTELLIGENCE SYSTEM FOR PHYTOCHEMICAL

PREDICTION AND PERSONALIZED PLANT RECOMMENDATION

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THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES

THE INVENTION AND THE MANNER IN WHICH IT IS TO BE

PERFORMED

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to machine learning systems for pharmaceutical informatics and geoinformatics, and more particularly to an
10 integrated geo-medicinal intelligence system that combines location-aware phytochemical concentration prediction, personalized bioavailability estimation, herb-drug interaction inference, multi-objective plant recommendation, and spectral image-based compound quantification for medicinal plants.

BACKGROUND

15 [0004] The following text of background art is provided for purposes of understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] Medicinal plants represent a valuable source of bioactive compounds for therapeutic applications. However, the effective utilization of these plants in
20 clinical settings faces several challenges related to variability, personalization, safety, and quality assessment.

[0006] Phytochemical concentrations in medicinal plants vary based on geographic location, elevation, climate, and soil composition. Existing medicinal plant databases typically provide static species monographs with single fixed values
25 for phytochemical content, ignoring this geographic and environmental variation. CN-113016508-B discloses a medicinal plant cultivation AI method that predicts medicinal efficacy based on environmental conditions including annual rainfall, soil characteristics, terrain conditions, and light intensity using machine learning models. However, such approaches may not provide uncertainty quantification or
30 confidence intervals that enable risk-aware clinical decision-making.

- [0007] Dosage recommendations for herbal medicines are often based on population-average estimates without accounting for individual patient characteristics such as age, weight, organ function, and other physiological parameters. CN-114550847-B discloses a method for predicting drug oral bioavailability using graph convolutional neural networks with molecular descriptors and SMILES expressions as inputs. While this approach addresses bioavailability prediction at the molecular level, it may not incorporate patient-specific demographic and health indicators for personalized dosage estimation.
- [0008] Herb-drug interactions present safety concerns in clinical practice, particularly when patients combine herbal medicines with pharmaceutical drugs. Current herb-drug interaction databases rely on manual curation, leaving undocumented interactions invisible to practitioners. IN-202511076502-A discloses a system for predicting synergistic herbal combinations using graph neural networks to model herb-target-disease relationships. However, such systems may focus on herb-herb synergies rather than herb-drug interaction prediction for previously undocumented compound pairs.
- [0009] Clinical decision support systems for traditional medicine have been developed to assist practitioners in treatment selection. IN-202521018486-A discloses an AI-based clinical decision support system for Ayurvedic drug dosage recommendations that analyzes patient data using machine learning algorithms including decision trees, random forests, and deep learning models. Such systems may provide dosage guidance but may not address the inverse problem of recommending plants based on clinical conditions, patient profiles, and geographic constraints.
- [0010] Quality assurance for medicinal plant materials typically requires laboratory testing using techniques such as high-performance liquid chromatography (HPLC) or gas chromatography-mass spectrometry (GC-MS). These methods, while accurate, may be time-consuming and costly, making them less suitable for field-scale deployment and real-time quality assessment.

[0011] Accordingly, there exists a need for improved systems and methods that address one or more of these challenges in medicinal plant analysis and clinical decision support.

[0012] The present invention addresses the above shortcomings of the prior art.

5 However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

SUMMARY

[0013] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of
10 the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

15 [0014] According to an aspect of the present disclosure, a geo-medicinal intelligence system is provided. The system includes a geographic potency prediction module configured to predict phytochemical concentrations for a medicinal plant at a specified geographic location. The geographic potency prediction module receives geographic location input and harvest month data. The
20 geographic potency prediction module retrieves environmental features for the location including temperature, rainfall, and soil properties. The geographic potency prediction module constructs a geo-environmental feature vector from the environmental features. The geographic potency prediction module applies Gaussian Process Regression with a radial basis function kernel to model non-linear
25 geo-environmental relationships. The geographic potency prediction module applies a seasonal kernel for temporal autocorrelation. The geographic potency prediction module computes predicted phytochemical concentrations per compound. The geographic potency prediction module computes confidence intervals from a Gaussian Process covariance matrix. The geographic potency
30 prediction module outputs predicted concentrations with uncertainty bounds.

[0015] The geographic potency prediction module provides the technical advantage of enabling location-aware phytochemical concentration prediction without requiring field measurement at each location. The uncertainty quantification through confidence intervals enables risk-aware clinical decision-making that is unavailable in static plant databases providing single fixed values. The combination of spatial and temporal modeling through the seasonal kernel captures both geographic and seasonal variation in phytochemical content.

[0016] According to other aspects of the present disclosure, the system may include one or more of the following features. The geo-environmental feature vector may include latitude, longitude, altitude, temperature, rainfall, soil pH, nitrogen content, phosphorus content, and potassium content. The seasonal kernel may comprise a product of a radial basis function kernel and a periodic kernel. The confidence intervals may comprise 95% confidence intervals.

[0017] These features provide the technical advantage of capturing the complex interplay between multiple environmental factors that influence phytochemical biosynthesis in medicinal plants. The periodic kernel component enables modeling of cyclical seasonal patterns in compound production.

[0018] According to another aspect of the present disclosure, a personalized dosage estimation module is provided. The personalized dosage estimation module is configured to estimate patient-specific bioavailability for phytochemical compounds. The personalized dosage estimation module receives patient profile features including age, weight, gender, body mass index, kidney function, and liver function. The personalized dosage estimation module receives compound descriptors including molecular weight, logP, hydrogen bond acceptors, and hydrogen bond donors. The personalized dosage estimation module receives a delivery mode. The personalized dosage estimation module feeds combined features into a Bayesian Neural Network. The personalized dosage estimation module performs Monte Carlo dropout sampling at inference. The personalized dosage estimation module computes a posterior distribution of patient-specific bioavailability. The personalized dosage estimation module derives an effective dose range from the posterior distribution. The personalized dosage estimation

module outputs a bioavailability estimate and personalized dosage range with uncertainty.

5 **[0019]** The personalized dosage estimation module provides the technical advantage of generating patient-specific dosage recommendations rather than population-average estimates. The Bayesian Neural Network architecture with Monte Carlo dropout enables probabilistic outputs suitable for clinical risk assessment. The integration of patient demographic features, compound descriptors, and delivery mode enables personalization across multiple dimensions simultaneously.

10 **[0020]** According to other aspects of the present disclosure, the personalized dosage estimation module may include one or more of the following features. The patient profile features may include twelve or more demographic and health indicators. The delivery mode may include oral, sublingual, or transdermal administration routes. The effective dose range may be expressed as a mean value
15 plus or minus a standard deviation.

[0021] These features provide the technical advantage of accommodating diverse patient populations and administration methods within a unified prediction framework. The probabilistic dose range output enables practitioners to select dosages appropriate for individual patient risk profiles.

20 **[0022]** According to another aspect of the present disclosure, an herb-drug interaction prediction module is provided. The herb-drug interaction prediction module is configured to predict interaction probability for herb-drug pairs including previously undocumented combinations. The herb-drug interaction prediction module receives a candidate herb-drug pair. The herb-drug interaction prediction
25 module retrieves or constructs a knowledge graph containing known herb-drug interactions. The herb-drug interaction prediction module trains a Graph Neural Network to learn node embeddings from the knowledge graph. The herb-drug interaction prediction module extracts graph embeddings for the herb and drug. The herb-drug interaction prediction module computes structural features for the pair
30 including fingerprint similarity, pathway overlap, and metabolic markers. The herb-drug interaction prediction module feeds combined graph embeddings and

structural features into a multilayer perceptron classifier. The herb-drug interaction prediction module computes an interaction probability. The herb-drug interaction prediction module classifies a risk category based on a probability threshold. The herb-drug interaction prediction module outputs the interaction probability and risk category.

[0023] The herb-drug interaction prediction module provides the technical advantage of inferring interactions for previously undocumented herb-drug combinations through learned embedding space representations. This addresses the limitation of manually curated interaction databases that leave undocumented interactions invisible to practitioners. The combination of graph-based embeddings and structural features enables generalization to novel compound pairs not present in training data.

[0024] According to other aspects of the present disclosure, the herb-drug interaction prediction module may include one or more of the following features.

The knowledge graph may include nodes representing medicinal plants, pharmaceutical drugs, compounds, conditions, and targets. The node embeddings may comprise 64-dimensional learned representations. The risk category may include low, moderate, or high classifications.

[0025] These features provide the technical advantage of capturing rich semantic relationships between diverse entity types in the biomedical domain. The multi-dimensional embedding space enables nuanced similarity comparisons between entities.

[0026] According to another aspect of the present disclosure, a smart recommendation engine is provided. The smart recommendation engine is configured to generate ranked medicinal plant recommendations based on clinical condition, patient profile, and geographic location. The smart recommendation engine receives a clinical condition, patient profile, and geographic location as input constraints. The smart recommendation engine retrieves a candidate plant set from a database. The smart recommendation engine computes a potency match score for each candidate plant using the geographic potency prediction module. The smart recommendation engine computes a safety score using the herb-drug

interaction prediction module. The smart recommendation engine computes a personalization fit score using the personalized dosage estimation module. The smart recommendation engine computes an availability score based on plant availability at the specified location. The smart recommendation engine applies multi-objective constraint optimization across the potency match score, safety score, personalization fit score, and availability score. The smart recommendation engine ranks candidate plants by an integrated score. The smart recommendation engine outputs ranked plant recommendations with per-dimension scores, recommended dosages, and confidence levels.

10 **[0027]** The smart recommendation engine provides the technical advantage of solving the inverse problem of recommending plants based on clinical conditions rather than merely providing information about what a given plant treats. The multi-objective optimization simultaneously balances potency, safety, personalization, and availability considerations. The integration of outputs from multiple specialized modules enables holistic recommendation generation.

[0028] According to other aspects of the present disclosure, the smart recommendation engine may include one or more of the following features. The multi-objective constraint optimization may apply weighted linear aggregation of the individual scores. The constraints may include safety thresholds, feasible dosage ranges, and regulatory restrictions. The output may include confidence metrics for each recommendation.

20 **[0029]** These features provide the technical advantage of enabling customizable weighting of different optimization objectives based on clinical priorities. The constraint satisfaction ensures that recommendations meet minimum safety and feasibility criteria.

[0030] According to another aspect of the present disclosure, a spectral image analyzer module is provided. The spectral image analyzer module is configured to quantify phytochemical compound concentrations from spectral images of plant samples. The spectral image analyzer module captures a multi-channel spectral image of a plant sample. The spectral image analyzer module pre-processes the spectral image tensor. The spectral image analyzer module applies a first

convolutional layer with batch normalization and ReLU activation. The spectral image analyzer module applies max pooling for spatial down-sampling. The spectral image analyzer module applies a second convolutional layer with batch normalization and ReLU activation. The spectral image analyzer module applies global average pooling. The spectral image analyzer module passes features through a dense layer with dropout. The spectral image analyzer module passes features through an output layer. The spectral image analyzer module outputs estimated compound concentrations for multiple phytochemical classes.

[0031] The spectral image analyzer module provides the technical advantage of enabling rapid, non-destructive phytochemical quantification in field settings without requiring laboratory equipment. The processing time of approximately 0.14 seconds compared to approximately 36 hours for laboratory analysis enables real-time quality assessment. The reduced cost per sample enables field-scale deployment for quality control applications.

[0032] According to other aspects of the present disclosure, the spectral image analyzer module may include one or more of the following features. The multi-channel spectral image may comprise a five-channel image including RGB-equivalent bands and near-infrared and thermal bands. The spectral image tensor may have dimensions of 32 by 32 pixels by 5 channels. The phytochemical classes may include alkaloids, flavonoids, terpenoids, and glycosides. The first convolutional layer may include 32 filters with a 3 by 3 kernel size. The second convolutional layer may include 64 filters with a 3 by 3 kernel size. The dense layer may include 128 units with dropout of 0.3.

[0033] These features provide the technical advantage of capturing biochemical content information through near-infrared and thermal spectral bands beyond visible light. The convolutional neural network architecture enables automatic feature extraction from raw spectral data without manual feature engineering.

[0034] According to another aspect of the present disclosure, an integrated geo-medicinal intelligence system is provided. The integrated system includes the geographic potency prediction module, the personalized dosage estimation module,

the herb-drug interaction prediction module, the smart recommendation engine, and the spectral image analyzer module. The integrated system includes an orchestration layer configured to receive user input including geographic location, patient profile, clinical condition, and an optional spectral image. The orchestration layer calls the spectral image analyzer module if a spectral image is provided. The orchestration layer calls the geographic potency prediction module to predict phytochemical concentrations. The orchestration layer calls the personalized dosage estimation module to estimate patient-specific bioavailability. The orchestration layer calls the herb-drug interaction prediction module to predict interaction risks. The orchestration layer calls the smart recommendation engine to generate plant recommendations. The orchestration layer compiles outputs from all modules into a structured decision report including recommended plants, potency scores, personalized dosages, interaction warnings, and confidence metrics.

[0035] The integrated system provides the technical advantage of combining location-aware predictions, personalized dosing, interaction checking, reverse recommendations, and spectral analysis in one unified tool. The end-to-end pipeline from input to structured decision report enables practical deployment for clinical decision support. The integration of uncertainty quantification across modules enables risk-aware decision-making throughout the recommendation process.

[0036] According to another aspect of the present disclosure, a method for operating a geo-medicinal intelligence system is provided. The method includes receiving user input including geographic location, patient profile, clinical condition, and an optional spectral image. The method includes analyzing the spectral image using a spectral image analyzer module if a spectral image is provided. The method includes predicting geographic potency of phytochemical compounds using a geographic potency prediction module. The method includes estimating personalized dosage using a personalized dosage calculator module. The method includes predicting herb-drug interactions using an interaction prediction module. The method includes generating plant recommendations using a smart recommendation engine that aggregates outputs from all modules. The method includes outputting a ranked decision report including recommended plants,

potency scores, personalized dosages, interaction warnings, and confidence metrics.

[0037] The method provides the technical advantage of enabling systematic integration of multiple analytical dimensions into a unified recommendation workflow. The sequential or parallel execution of specialized modules enables efficient processing while maintaining comprehensive analysis coverage.

[0038] According to another aspect of the present disclosure, a method for geographic potency prediction is provided. The method includes receiving geographic location input and harvest month. The method includes retrieving environmental features for the location including temperature, rainfall, and soil properties. The method includes constructing a geo-environmental feature vector. The method includes applying Gaussian Process Regression with a radial basis function kernel to model non-linear geo-environmental relationships. The method includes applying a seasonal kernel for temporal autocorrelation. The method includes computing predicted phytochemical concentrations per compound. The method includes computing confidence intervals from a Gaussian Process covariance matrix. The method includes outputting predicted concentrations with uncertainty bounds.

[0039] The method provides the technical advantage of enabling prediction of phytochemical potency for any geographic location without requiring direct field measurement at that location. The Gaussian Process framework provides principled uncertainty quantification unavailable in deterministic prediction methods.

[0040] According to another aspect of the present disclosure, a method for personalized dosage estimation is provided. The method includes receiving patient profile features including age, weight, gender, body mass index, kidney function, and liver function. The method includes receiving compound descriptors including molecular weight, logP, hydrogen bond acceptors, and hydrogen bond donors. The method includes receiving a delivery mode. The method includes feeding combined features into a Bayesian Neural Network. The method includes performing Monte Carlo dropout sampling at inference. The method includes computing a posterior distribution of patient-specific bioavailability. The method includes deriving an

effective dose range from the posterior distribution. The method includes outputting a bioavailability estimate and personalized dosage range with uncertainty.

[0041] The method provides the technical advantage of generating probabilistic dosage estimates that account for individual patient variability rather than relying on population-average values. The uncertainty quantification enables clinical practitioners to make risk-informed dosing decisions.

[0042] According to another aspect of the present disclosure, a method for herb-drug interaction prediction is provided. The method includes receiving a candidate herb-drug pair. The method includes retrieving or constructing a knowledge graph containing known herb-drug interactions. The method includes training a Graph Neural Network to learn node embeddings. The method includes extracting graph embeddings for the herb and drug. The method includes computing structural features for the pair including fingerprint similarity, pathway overlap, and metabolic markers. The method includes feeding combined features into a multilayer perceptron classifier. The method includes computing an interaction probability. The method includes classifying a risk category based on a probability threshold. The method includes outputting the interaction probability and risk category.

[0043] The method provides the technical advantage of enabling prediction of interactions for herb-drug combinations not present in existing databases through learned embedding representations. The combination of graph-based and structural features enables robust generalization to novel compound pairs.

[0044] According to another aspect of the present disclosure, a method for smart plant recommendation is provided. The method includes receiving a clinical condition, patient profile, and geographic location as input constraints. The method includes retrieving a candidate plant set from a database. The method includes computing a potency match score for each candidate plant using a geographic potency prediction module. The method includes computing a safety score using an interaction prediction module. The method includes computing a personalization fit score using a personalized dosage calculator module. The method includes computing an availability score based on plant availability at the specified location.

The method includes applying multi-objective constraint optimization. The method includes ranking candidate plants by an integrated score. The method includes outputting ranked plant recommendations with per-dimension scores, recommended dosages, and confidence levels.

5 **[0045]** The method provides the technical advantage of addressing the inverse problem of deriving optimal medicinal plants from condition constraints rather than merely providing forward lookup of plant properties. The multi-objective optimization enables balanced consideration of competing clinical priorities.

10 **[0046]** According to another aspect of the present disclosure, a method for spectral image compound quantification is provided. The method includes capturing a multi-channel spectral image of a plant sample. The method includes pre-processing the spectral image tensor. The method includes applying a first convolutional layer with batch normalization and ReLU activation. The method includes applying max pooling for spatial down-sampling. The method includes
15 applying a second convolutional layer with batch normalization and ReLU activation. The method includes applying global average pooling. The method includes passing features through a dense layer with dropout. The method includes passing features through an output layer. The method includes outputting estimated compound concentrations for multiple phytochemical classes.

20 **[0047]** The method provides the technical advantage of enabling rapid phytochemical quantification suitable for field deployment without requiring laboratory infrastructure. The convolutional neural network architecture enables end-to-end learning from raw spectral data to compound concentration estimates.

25 **[0048]** The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

30 **[0049]** The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed

description, depict the operational flow, component architecture, and other mechanisms.

[0050] FIG. 1 illustrates a flowchart of a method for operating a geo-medicinal intelligence system, according to aspects of the present disclosure.

5 **[0051]** FIG. 2 illustrates a flowchart of a method for geographic potency prediction, according to aspects of the present disclosure.

[0052] FIG. 3 illustrates a flowchart of a method for personalized dosage estimation, according to an embodiment.

10 **[0053]** FIG. 4 illustrates a flowchart of a method for herb-drug interaction prediction, according to aspects of the present disclosure.

[0054] FIG. 5 illustrates a flowchart of a method for smart plant recommendation, according to an embodiment.

[0055] FIG. 6 illustrates a flowchart of a method for spectral image compound quantification, according to aspects of the present disclosure.

15 **[0056]** FIG. 7 illustrates geographic potency prediction results for turmeric alkaloid distribution, according to an embodiment.

[0057] FIG. 8 illustrates performance characteristics of a personalized bioavailability estimation module, according to aspects of the present disclosure.

20 **[0058]** FIG. 9 illustrates a drug-herb interaction knowledge graph and associated performance metrics, according to an embodiment.

[0059] FIG. 10 illustrates recommendation system outputs for plant-condition pairings, according to aspects of the present disclosure.

25 **[0060]** FIG. 11 illustrates training and performance characteristics of a convolutional neural network for spectral compound quantification, according to an embodiment.

[0061] FIG. 12 illustrates recommendation system outputs for multiple patient profiles, according to aspects of the present disclosure.

[0062] FIG. 13 illustrates relationships between alkaloid concentration and geographic variables, according to an embodiment.

[0063] FIG. 14 illustrates a performance evaluation comparing a Gaussian Process system against baseline methods, according to aspects of the present disclosure.

5 [0064] FIG. 15 illustrates performance of a Bayesian Neural Network for bioavailability prediction compared to baseline methods, according to an embodiment.

[0065] FIG. 16 illustrates a performance evaluation for an herb-drug interaction prediction system, according to aspects of the present disclosure.

10 [0066] FIG. 17 illustrates performance evaluation of a recommendation system compared to baseline approaches, according to an embodiment.

[0067] FIG. 18 illustrates performance comparison between a convolutional neural network and laboratory measurements for compound quantification, according to aspects of the present disclosure.

15 [0068] FIG. 19 illustrates a competitive advantage summary for a geo-medicinal intelligence system, according to an embodiment.

[0069] FIG. 20 illustrates validation metrics for Bayesian calibration and model performance, according to aspects of the present disclosure.

[0070] FIG. 21 illustrates relationships between alkaloid concentration and geographic or environmental variables, according to an embodiment.

20 DETAILED DESCRIPTION

[0071] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

25 [0072] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements.

30 Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

[0073] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0074] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0075] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated by the same reference signs.

[0076] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0077] FIG. 1 illustrates a method 100 for operating a geo-medicinal intelligence system according to various embodiments. Method 100 describes the operational flow of an integrated system that combines multiple specialized computational modules to generate comprehensive medicinal plant recommendations based on geographic, patient-specific, and clinical inputs.

[0078] Method 100 includes a step 102 of receiving user input. In step 102, the system receives input data including a geographic location, a patient profile, a

clinical condition, and an optional spectral image. The geographic location may include latitude, longitude, and altitude coordinates corresponding to a region where medicinal plants are sourced or where the patient is located. The patient profile may include demographic and health indicators such as age, weight, gender, body mass index, kidney function, and liver function. The clinical condition may specify a disease state, symptom set, or therapeutic objective for which medicinal plant recommendations are sought. The optional spectral image may comprise a multi-channel image of a plant sample captured using a spectral imaging device.

[0079] Step 102 proceeds to a step 104 of analyzing the spectral image using a spectral image analyzer module. In step 104, if a spectral image is provided as part of the user input, the spectral image analyzer module processes the spectral image to quantify phytochemical compound concentrations in the plant sample. The spectral image analyzer module may apply convolutional neural network layers to extract features from the spectral image tensor and output estimated concentrations for multiple phytochemical classes including alkaloids, flavonoids, terpenoids, and glycosides. In some embodiments, step 104 may be bypassed if no spectral image is provided in step 102.

[0080] Step 104 proceeds to a step 106 of predicting geographic potency of phytochemical compounds using a geographic potency prediction module. In step 106, the geographic potency prediction module receives the geographic location and harvest month data from the user input. The geographic potency prediction module retrieves environmental features for the specified location including temperature, rainfall, and soil properties. The geographic potency prediction module constructs a geo-environmental feature vector and applies Gaussian Process Regression with a radial basis function kernel to model non-linear relationships between environmental factors and phytochemical concentrations. The geographic potency prediction module applies a seasonal kernel for temporal autocorrelation and computes predicted phytochemical concentrations per compound along with confidence intervals from a Gaussian Process covariance matrix. The output of step 106 includes predicted concentrations with uncertainty bounds that enable risk-aware clinical decision-making.

[0081] Step 106 proceeds to a step 108 of estimating personalized dosage using a personalized dosage calculator module. In step 108, the personalized dosage calculator module receives patient profile features from the user input including age, weight, gender, body mass index, kidney function, and liver function. The personalized dosage calculator module also receives compound descriptors including molecular weight, logP, hydrogen bond acceptors, and hydrogen bond donors for candidate phytochemical compounds. The personalized dosage calculator module receives a delivery mode specifying an administration route. The personalized dosage calculator module feeds combined features into a Bayesian Neural Network and performs Monte Carlo dropout sampling at inference to compute a posterior distribution of patient-specific bioavailability. The personalized dosage calculator module derives an effective dose range from the posterior distribution and outputs a bioavailability estimate and personalized dosage range with uncertainty.

[0082] Step 108 proceeds to a step 110 of predicting herb-drug interactions using an interaction prediction module. In step 110, the interaction prediction module receives candidate herb-drug pairs based on the patient profile and any concurrent medications. The interaction prediction module retrieves or constructs a knowledge graph containing known herb-drug interactions. The interaction prediction module trains a Graph Neural Network to learn node embeddings from the knowledge graph and extracts graph embeddings for each herb and drug in the candidate pairs. The interaction prediction module computes structural features for each pair including fingerprint similarity, pathway overlap, and metabolic markers. The interaction prediction module feeds combined graph embeddings and structural features into a multilayer perceptron classifier to compute an interaction probability and classify a risk category based on a probability threshold. The output of step 110 includes interaction probabilities and risk categories for each candidate herb-drug pair.

[0083] Step 110 proceeds to a step 112 of generating plant recommendations using a smart recommendation engine. In step 112, the smart recommendation engine aggregates outputs from the geographic potency prediction module, the

personalized dosage calculator module, and the interaction prediction module. The smart recommendation engine receives the clinical condition, patient profile, and geographic location as input constraints and retrieves a candidate plant set from a database. For each candidate plant, the smart recommendation engine computes a
5 potency match score using the geographic potency prediction module output, a safety score using the interaction prediction module output, a personalization fit score using the personalized dosage calculator module output, and an availability score based on plant availability at the specified location. The smart recommendation engine applies multi-objective constraint optimization across the
10 potency match score, safety score, personalization fit score, and availability score. The smart recommendation engine ranks candidate plants by an integrated score that balances the multiple optimization objectives.

[0084] Step 112 proceeds to a step 114 of outputting a ranked decision report. In step 114, the system compiles outputs from all modules into a structured decision
15 report. The ranked decision report includes recommended plants listed in order of integrated score, potency scores indicating predicted phytochemical concentrations at the specified geographic location, personalized dosages tailored to the patient profile, interaction warnings identifying herb-drug combinations with elevated risk categories, and confidence metrics quantifying uncertainty across the various
20 predictions. The ranked decision report provides a comprehensive output that enables clinical practitioners to make informed decisions regarding medicinal plant selection and dosing.

[0085] The data flow between modules in method 100 enables integration of multiple analytical dimensions into a unified recommendation workflow. The
25 geographic potency prediction module provides location-specific phytochemical concentration estimates that inform both the potency match score computation in the smart recommendation engine and the compound descriptor inputs for the personalized dosage calculator module. The personalized dosage calculator module provides patient-specific bioavailability estimates that inform the personalization
30 fit score computation in the smart recommendation engine. The interaction prediction module provides safety assessments that inform the safety score

computation in the smart recommendation engine. The spectral image analyzer module, when utilized, provides measured compound concentrations that may supplement or validate the predicted concentrations from the geographic potency prediction module. The smart recommendation engine aggregates outputs from all modules to generate ranked recommendations that simultaneously account for potency, safety, personalization, and availability considerations.

[0086] FIG. 2 illustrates a method 200 for geographic potency prediction according to various embodiments. Method 200 describes the operational flow of a geographic potency prediction module that predicts phytochemical concentrations for medicinal plants at specified geographic locations using Gaussian Process Regression with seasonal kernel adaptation.

[0087] Method 200 includes a step 202 of receiving geographic location input and harvest month. In step 202, the geographic potency prediction module receives input data specifying a geographic location where a medicinal plant is sourced or where phytochemical concentration prediction is desired. The geographic location input may include latitude coordinates, longitude coordinates, and altitude values. The harvest month input specifies a temporal parameter corresponding to the month during which the medicinal plant is harvested or is expected to be harvested. The combination of geographic location and harvest month enables the geographic potency prediction module to account for both spatial and temporal variation in phytochemical content.

[0088] Step 202 proceeds to a step 204 of retrieving environmental features for the location including temperature, rainfall, and soil properties. In step 204, the geographic potency prediction module retrieves environmental data corresponding to the specified geographic location. The environmental features may include temperature values representing average or seasonal temperature at the location. The environmental features may include rainfall values representing precipitation levels at the location. The environmental features may include soil properties such as soil pH, nitrogen content, phosphorus content, and potassium content. In some embodiments, the environmental features may be retrieved from environmental databases, satellite data sources, or climate monitoring systems.

[0089] Step 204 proceeds to a step 206 of constructing a geo-environmental feature vector. In step 206, the geographic potency prediction module constructs a feature vector that combines the geographic location parameters and the environmental features into a unified representation. The geo-environmental feature vector may include latitude, longitude, altitude, temperature, rainfall, soil pH, nitrogen content, phosphorus content, and potassium content. In some embodiments, the geo-environmental feature vector may include 28 or more geo-environmental features combined with compound descriptors. The geo-environmental feature vector serves as input to the Gaussian Process Regression model for phytochemical concentration prediction.

[0090] Step 206 proceeds to a step 208 of applying Gaussian Process Regression with a radial basis function (RBF) kernel to model non-linear geo-environmental relationships. In step 208, the geographic potency prediction module applies Gaussian Process Regression to the geo-environmental feature vector. The Gaussian Process Regression model uses a radial basis function kernel to capture non-linear relationships between environmental factors and phytochemical concentrations. The radial basis function kernel enables the model to learn complex spatial patterns in phytochemical distribution that may not be captured by linear regression methods. The Gaussian Process Regression model may be expressed as:

$$f(x_{geo}) \sim \mathcal{GP}(\mu(x_{geo}), K_{RBF}(x_{geo}, x'_{geo}) \cdot K_{seasonal}(t, t'))$$

[0091] where x_{geo} represents the geo-environmental feature vector and $K_{seasonal}$ represents the seasonal kernel.

[0092] Step 208 proceeds to a step 210 of applying a seasonal kernel for temporal autocorrelation. In step 210, the geographic potency prediction module applies a seasonal kernel component to capture temporal patterns in phytochemical production. The seasonal kernel may comprise a product of a radial basis function kernel and a periodic kernel, expressed as:

$$K_{seasonal}(t, t') = K_{RBF}(t, t') \cdot K_{periodic}(t, t')$$

$$K_{periodic}(t, t') = \sigma^2 \exp \left(- \frac{2 \sin^2(\pi |t - t'|/p)}{l^2} \right)$$

[0093] The periodic kernel component enables the model to capture cyclical seasonal patterns in compound production that repeat on an annual basis. The combination of the radial basis function kernel and the periodic kernel enables the geographic potency prediction module to model both geographic variation and seasonal variation in phytochemical concentrations simultaneously.

[0094] Step 210 proceeds to a step 212 of computing predicted phytochemical concentrations per compound. In step 212, the geographic potency prediction module computes predicted concentrations for multiple phytochemical compounds based on the Gaussian Process Regression model output. The predicted concentrations may include alkaloid concentrations, flavonoid concentrations, terpenoid concentrations, and glycoside concentrations expressed as percentage of dry weight. The geographic potency prediction module may compute predictions for multiple compounds simultaneously using the same geo-environmental feature vector input.

[0095] Step 212 proceeds to a step 214 of computing 95% confidence intervals from a Gaussian Process covariance matrix. In step 214, the geographic potency prediction module computes confidence intervals that quantify uncertainty in the predicted phytochemical concentrations. The confidence intervals are derived from the Gaussian Process covariance matrix, which provides a measure of prediction variance at each location. In some embodiments, the confidence intervals comprise 95% confidence intervals. The uncertainty quantification enables risk-aware clinical decision-making by providing practitioners with information about the reliability of predicted concentrations at different geographic locations.

[0096] Step 214 proceeds to a step 216 of outputting predicted concentrations with uncertainty bounds. In step 216, the geographic potency prediction module outputs the predicted phytochemical concentrations along with the computed confidence intervals. The output may be expressed as a mean predicted concentration value plus or minus a standard deviation or confidence interval width. The output enables downstream modules such as the smart recommendation engine to incorporate location-specific potency information with quantified uncertainty into plant recommendation decisions.

[0097] FIG. 7 illustrates geographic potency prediction results for turmeric alkaloid distribution according to various embodiments. FIG. 7 presents three visualization panels that demonstrate the geographic potency prediction capabilities across spatial and temporal dimensions. A first panel of FIG. 7 presents a three-dimensional scatter plot showing the relationship between latitude, longitude, and altitude coordinates for turmeric sampling locations. The latitude values range from approximately 10 to 30 degrees, the longitude values range from approximately 70 to 95 degrees, and the altitude values range from 0 to approximately 2000 meters, representing the geographic spread of turmeric sampling locations across Indian regions. A second panel of FIG. 7 displays a predicted versus actual comparison plot with an R-squared value of 0.839, demonstrating the model predictive accuracy for alkaloid concentrations. The actual alkaloid percentage values range from approximately 2 to 5 percent, and the data points cluster closely around a diagonal reference line representing perfect prediction. A third panel of FIG. 7 illustrates monthly alkaloid variation in the Nilgiris region, showing temporal fluctuations in alkaloid content throughout the year. The alkaloid percentage ranges from approximately 2.5 to 4.5 percent across months 1 to 12, with a shaded region representing the 95% confidence interval demonstrating the uncertainty quantification capability of the prediction system.

[0098] FIG. 13 illustrates relationships between alkaloid concentration and geographic variables from a NAPRALERT-style dataset according to various embodiments. FIG. 13 presents three data visualization panels. A first panel of FIG. 13 displays a scatter plot showing alkaloid concentration versus latitude, with data points for multiple plant species plotted against latitude on the horizontal axis and alkaloid percentage on the vertical axis. The data points cluster between latitudes of approximately 10 to 35 degrees and alkaloid concentrations ranging from approximately 0.5 to 2.5 percent. A second panel of FIG. 13 shows a horizontal bar chart displaying the number of records collected from various Indian geographic locations including Bhopal in Madhya Pradesh, Pune in Maharashtra, Tirupati in Andhra Pradesh, Bhubaneswar in Odisha, Madurai in Tamil Nadu, Ranchi in Jharkhand, Nauni in Himachal Pradesh, Junagadh in Gujarat, Ahmedabad in

Gujarat, and Chandigarh in Punjab, with record counts ranging from approximately 100 to 175 samples per region. A third panel of FIG. 13 presents a scatter plot showing the relationship between temperature and alkaloid percentage, with data points color-coded according to altitude ranging from approximately 0 to 2000 meters. The temperature values range from approximately 15 to 35 degrees Celsius.

[0099] FIG. 14 illustrates a performance evaluation comparing a Gaussian Process system against baseline methods according to various embodiments. FIG. 14 presents a performance evaluation dashboard comparing the Gaussian Process system against Linear Regression, K-Nearest Neighbors (KNN), and Random Forest baseline methods. A first panel of FIG. 14 displays a predicted versus actual scatter plot for the Gaussian Process system, showing data points for multiple compounds including alkaloids, flavonoids, terpenoids, and glycosides clustered around a diagonal reference line. A second panel of FIG. 14 shows a root mean square error (RMSE) comparison bar chart. A third panel of FIG. 14 presents an R-squared comparison with a target threshold of 0.80 marked by a reference line. A fourth panel of FIG. 14 displays error distribution box plots by system. A fifth panel of FIG. 14 shows per-compound RMSE values as grouped bar charts for alkaloids, flavonoids, terpenoids, and glycosides across all four methods. A sixth panel of FIG. 14 contains a capability comparison matrix evaluating each system across criteria including uncertainty quantification, temporal variation handling, multi-compound prediction capability, and real-time operation, with the Gaussian Process system showing capability for uncertainty quantification and temporal variation handling.

[0100] FIG. 21 illustrates relationships between alkaloid concentration and geographic or environmental variables according to various embodiments. FIG. 21 comprises two scatter plots. A first scatter plot of FIG. 21 displays alkaloid percentage on the vertical axis ranging from approximately 1.5 to 4.5 percent, plotted against latitude in degrees North on the horizontal axis spanning from approximately 10 to 30 degrees. Data points are color-coded to represent three different plant species: Turmeric, Ashwagandha, and Gymnema. The scatter distribution shows variability in alkaloid concentration across the latitude range for

all three species. A second scatter plot of FIG. 21 presents alkaloid percentage on the vertical axis against temperature in degrees Celsius on the horizontal axis, ranging from approximately 10 to 40 degrees Celsius. The data points are colored according to altitude, with a gradient scale indicating elevation values from approximately 500 to 2500 meters. The distribution demonstrates the interplay between temperature, altitude, and phytochemical content.

[0101] Working Example 1: Geographic Potency Prediction Validation

[0102] The geographic potency prediction module was validated using a dataset comprising 1,474 literature-calibrated records across 15 Indian medicinal plants and 15 ecologically diverse Indian geographic regions. The plant species included Turmeric, Ashwagandha, Gymnema, Moringa, and Brahmi, among others. The geographic regions included the Western Ghats, Indo-Gangetic Plains, and Eastern Highlands, among others. The dataset included 28 geo-environmental features combined with compound descriptors. The target compounds included alkaloids, flavonoids, terpenoids, and glycosides expressed as percentage of dry weight. The validation strategy employed 5-fold cross-validation with spatial holdout, where test regions were unseen during training.

[0103] Table 1 presents the phytochemical concentration ranges observed in the dataset:

Compound	Range (% dry weight)	Mean	Standard Deviation
Alkaloids	0.12 - 2.45%	0.89%	0.34%
Flavonoids	0.23 - 4.67%	1.98%	0.92%
Terpenoids	0.15 - 3.89%	1.67%	0.71%
Glycosides	0.05 - 1.23%	0.45%	0.18%

[0104] Table 2 presents the prediction performance metrics for the geographic potency prediction module:

Compound	RMSE (%)	R-squared	MAE (%)	95% CI Width
Alkaloids	0.619	0.016	0.38	±0.45

Flavonoids	0.827	0.115	0.52	±0.68
Terpenoids	0.408	0.219	0.24	±0.32
Glycosides	0.163	0.016	0.08	±0.11

[0105] The Gaussian Process model demonstrated an R-squared improvement of +0.18 on average compared to Linear Regression baseline. The Gaussian Process model demonstrated an R-squared improvement of +0.12 for flavonoids compared to Random Forest baseline. The Gaussian Process model provides uncertainty quantification through confidence interval bounds that is unavailable in deterministic prediction methods such as Linear Regression, KNN, and Random Forest.

[0106] Table 3 presents a comparison of the Gaussian Process system capabilities against baseline methods:

Capability	GP System	Linear Regression	KNN	Random Forest
Uncertainty Quantification	Yes	No	No	No
Temporal Variation Handling	Yes	No	No	No
Multi-Compound Prediction	Yes	Yes	Yes	Yes
Real-Time Operation	Yes	Yes	Yes	Yes

[0107] The geographic potency prediction module enables prediction of phytochemical potency for any geographic location without requiring direct field measurement at that location. The Gaussian Process framework provides principled uncertainty quantification that enables risk-aware clinical decision-making. The seasonal kernel adaptation captures both geographic and seasonal variation in

phytochemical content, addressing the limitation of static plant databases that provide single fixed values for phytochemical concentrations.

[0108] FIG. 3 illustrates a method 300 for personalized dosage estimation according to various embodiments. Method 300 describes the operational flow of a personalized dosage estimation module that estimates patient-specific bioavailability for phytochemical compounds using a Bayesian Neural Network architecture with Monte Carlo dropout sampling.

[0109] Method 300 includes a step 302 of receiving patient profile features. In step 302, the personalized dosage estimation module receives patient profile features including age, weight, gender, body mass index (BMI), kidney function, and liver function. The patient profile features may include 12 or more demographic and health indicators that characterize individual patient physiology. The patient profile features enable the personalized dosage estimation module to account for inter-patient variability in drug metabolism and absorption.

[0110] Step 302 proceeds to a step 304 of receiving compound descriptors. In step 304, the personalized dosage estimation module receives compound descriptors including molecular weight, logP, hydrogen bond acceptors (HBA), and hydrogen bond donors (HBD). The compound descriptors may include rotatable bonds and other molecular properties that influence bioavailability. The compound descriptors characterize the physicochemical properties of phytochemical compounds that affect absorption, distribution, metabolism, and excretion.

[0111] Step 304 proceeds to a step 306 of receiving delivery mode. In step 306, the personalized dosage estimation module receives a delivery mode specifying an administration route for the phytochemical compound. The delivery mode may include oral, sublingual, or transdermal administration routes. In some embodiments, the delivery mode may include infusion, decoction, powder, or extract formulations. The delivery mode affects bioavailability by influencing the absorption pathway and first-pass metabolism of the compound.

[0112] Step 306 proceeds to a step 308 of feeding combined features into a Bayesian Neural Network. In step 308, the personalized dosage estimation module combines the patient profile features, compound descriptors, and delivery mode

into a unified feature vector. The personalized dosage estimation module feeds the combined feature vector into a Bayesian Neural Network. The Bayesian Neural Network architecture enables probabilistic predictions that capture uncertainty in bioavailability estimates. The Bayesian Neural Network may be expressed as:

5
$$p(F | \mathbf{p}, \mathbf{c}, \mathbf{d}) = \int p(F | \mathbf{p}, \mathbf{c}, \mathbf{d}, W)p(W)dW$$

[0113] where $\mathbf{p}$ represents the patient profile features, $\mathbf{c}$ represents the compound descriptors, and $\mathbf{d}$ represents the delivery mode.

[0114] Step 308 proceeds to a step 310 of performing Monte Carlo dropout sampling at inference. In step 310, the personalized dosage estimation module performs Monte Carlo (MC) dropout sampling during inference to generate multiple stochastic forward passes through the Bayesian Neural Network. The Monte Carlo dropout sampling enables uncertainty quantification by sampling from the approximate posterior distribution of network weights. The multiple forward passes produce a distribution of bioavailability predictions rather than a single point estimate.

[0115] Step 310 proceeds to a step 312 of computing a posterior distribution of patient-specific bioavailability. In step 312, the personalized dosage estimation module aggregates the outputs from the Monte Carlo dropout sampling to compute a posterior distribution of patient-specific bioavailability. The posterior distribution captures both the expected bioavailability value and the uncertainty associated with the prediction. The posterior distribution enables risk-aware clinical decision-making by quantifying prediction confidence.

[0116] Step 312 proceeds to a step 314 of deriving an effective dose range from the posterior distribution. In step 314, the personalized dosage estimation module derives an effective dose range from the posterior distribution. The effective dose range may be expressed as a mean value plus or minus a standard deviation, represented as $[\mu - \sigma, \mu + \sigma]$. The effective dose range provides practitioners with a range of dosages appropriate for the individual patient rather than a single fixed value.

[0117] Step 314 proceeds to a step 316 of outputting a bioavailability estimate and personalized dosage range with uncertainty. In step 316, the personalized

dosage estimation module outputs the bioavailability estimate and personalized dosage range along with uncertainty quantification. The output includes the full posterior distribution of bioavailability, enabling downstream modules such as the smart recommendation engine to incorporate patient-specific dosing information with quantified uncertainty into plant recommendation decisions.

[0118] FIG. 8 illustrates performance characteristics of a personalized bioavailability estimation module according to various embodiments. FIG. 8 presents three visualization panels. A first panel of FIG. 8 displays training curves showing the convergence behavior of the Bayesian Neural Network over approximately 100 epochs, with both training loss and validation loss plotted against epoch number. The training curves demonstrate rapid initial descent followed by stabilization, indicating model convergence without overfitting as evidenced by close tracking between training and validation loss curves. A second panel of FIG. 8 presents a scatter plot comparing predicted bioavailability values against actual bioavailability measurements, with an R-squared value of 0.900. The data points cluster along a diagonal reference line representing perfect prediction accuracy, demonstrating strong correlation between model predictions and ground truth values across a bioavailability range of approximately 5 to 45 percent. A third panel of FIG. 8 displays a bar chart illustrating the effect of different delivery methods on bioavailability percentage. Four delivery methods are compared: infusion at approximately 20 percent bioavailability, decoction at approximately 15 percent bioavailability, powder at approximately 14 percent bioavailability, and extract at approximately 25 percent bioavailability. The extract delivery method shows the highest bioavailability, demonstrating how the personalized dosage estimation module accounts for delivery mode as a factor in personalized dosage calculations.

[0119] FIG. 15 illustrates performance of a Bayesian Neural Network for bioavailability prediction compared to baseline methods according to various embodiments. FIG. 15 presents four visualization panels. A first panel of FIG. 15 displays a scatter plot showing predicted bioavailability percentage against actual bioavailability percentage, where the Bayesian Neural Network achieves an R-

squared value of 0.901 with data points closely following a diagonal reference line, while a static baseline shows an R-squared of 0.001 with predictions clustered around a horizontal line near 20 percent. A second panel of FIG. 15 presents a bar chart comparing Mean Absolute Error (MAE) across four methods, where the
5 Bayesian Neural Network achieves an MAE of 1.97 percent, Random Forest achieves 1.88 percent, Age-Only Linear Regression shows 6.21 percent, and Static Mean baseline shows 6.34 percent. A third panel of FIG. 15 displays an R-squared comparison bar chart where the Bayesian Neural Network achieves 0.901, Random Forest achieves 0.915, Age-Only Linear Regression shows 0.033, and Static Mean
10 shows approximately 0.000. A fourth panel of FIG. 15 shows overlapping histograms of error distribution for both the Bayesian Neural Network and Static methods, with the Bayesian Neural Network distribution concentrated at lower absolute error values.

[0120] FIG. 20 illustrates validation metrics for Bayesian calibration and
15 model performance according to various embodiments. FIG. 20 presents two visualization panels. A first panel of FIG. 20 displays calibration coverage performance, showing a 95% confidence interval target at 95% and an achieved coverage at 89%. The achieved coverage of 89% demonstrates that the Bayesian Neural Network uncertainty quantification achieves calibration close to the
20 theoretical target, with predicted confidence intervals containing true values approximately 89% of the time. A second panel of FIG. 20 presents a model performance summary comparing ROC AUC scores across different model architectures, where the Bayesian Neural Network achieves an AUC of 0.960, exceeding a target threshold of 0.85.

25 **[0121] Working Example 2: Personalized Dosage Estimation Validation**

[0122] The personalized dosage estimation module was validated using a dataset comprising patient profiles with 12 demographic and health indicator features. The patient profile features included age, weight, gender, body mass index, kidney function, liver function, and additional physiological parameters. The
30 compound descriptors included molecular weight, logP, hydrogen bond acceptors,

hydrogen bond donors, and rotatable bonds. The delivery modes included oral, sublingual, transdermal, infusion, decoction, powder, and extract formulations.

[0123] Table 4 presents the bioavailability prediction performance metrics for the personalized dosage estimation module:

Model System	MAE (%)	R-squared	Personalizati on Capability	Inference Time
Bayesian NN (Proposed)	1.84	0.915	Full profile-aware (12 features) + uncertainty	8 ms
Random Forest Baseline	1.88	0.915	Multi-feature, deterministic	5 ms
Age-Only Linear	6.21	0.033	Partial (age only)	1 ms
Static Pharmacokinetic Tables	6.34	~0.000	None (population average)	<1 ms

- 5 [0124] The Bayesian Neural Network achieved an MAE of 1.84% compared to 6.34% for static pharmacokinetic tables, representing a reduction in error of approximately 70%. The Bayesian Neural Network achieved an R-squared of 0.915 compared to approximately 0.000 for static pharmacokinetic tables. The Bayesian Neural Network provides probabilistic outputs with full posterior distributions
- 10 rather than point estimates, enabling clinical risk assessment.

[0125] Table 5 presents the calibration coverage metrics for the Bayesian Neural Network uncertainty quantification:

Metric	Target	Achieved
95% Confidence Interval Coverage	95%	89%
ROC AUC Score	0.85	0.960

[0126] The achieved calibration coverage of 89% demonstrates that the Bayesian Neural Network uncertainty quantification approaches the theoretical 95% target, with predicted confidence intervals containing true values in approximately 89% of test cases. The ROC AUC score of 0.960 exceeds the target threshold of 0.85, validating the discriminative performance of the Bayesian Neural Network for personalized bioavailability estimation.

[0127] Table 6 presents a comparison of the Bayesian Neural Network system capabilities against baseline methods:

Capability	BNN System	Random Forest	Age-Only Linear	Static Tables
Uncertainty Quantification	Yes	No	No	No
Full Patient Profile	Yes	Yes	No	No
Compound Descriptors	Yes	Yes	No	No
Delivery Mode Adaptation	Yes	Yes	No	No
Probabilistic Output	Yes	No	No	No

[0128] The personalized dosage estimation module enables generation of patient-specific dosage recommendations rather than population-average estimates. The Bayesian Neural Network architecture with Monte Carlo dropout enables probabilistic outputs suitable for clinical risk assessment. The integration of patient demographic features, compound descriptors, and delivery mode enables personalization across multiple dimensions simultaneously, addressing the limitation of static pharmacokinetic tables that provide no personalization.

[0129] FIG. 4 illustrates a method 400 for herb-drug interaction prediction according to various embodiments. Method 400 describes the operational flow of

an herb-drug interaction prediction module that predicts interaction probability for herb-drug pairs including previously undocumented combinations using a Graph Neural Network combined with structural features.

[0130] Method 400 includes a step 402 of receiving a candidate herb-drug pair.

5 In step 402, the herb-drug interaction prediction module receives input data specifying a candidate herb-drug pair for interaction assessment. The candidate herb-drug pair may include a medicinal plant or herbal compound and a pharmaceutical drug that a patient may be taking concurrently. The herb-drug interaction prediction module may receive multiple candidate pairs for batch
10 processing when a patient is taking multiple medications.

[0131] Step 402 proceeds to a step 404 of retrieving or constructing a knowledge graph containing known herb-drug interactions. In step 404, the herb-drug interaction prediction module retrieves or constructs a knowledge graph that encodes known relationships between medicinal plants, pharmaceutical drugs,
15 compounds, conditions, and targets. The knowledge graph may include nodes representing 47 medicinal plants and 312 pharmaceutical drugs. The knowledge graph may include 1,847 edges representing known interactions between entities. The knowledge graph structure captures the topology of known interaction relationships that enables generalization to novel pairs through learned embeddings.

20 **[0132]** Step 404 proceeds to a step 406 of training a Graph Neural Network to learn node embeddings. In step 406, the herb-drug interaction prediction module trains a Graph Neural Network (GNN) on the knowledge graph to learn node embeddings that capture semantic relationships between entities. The Graph Neural Network learns to represent each node in the knowledge graph as a vector in a
25 learned embedding space. The node embeddings may comprise 64-dimensional learned representations. The Graph Neural Network architecture enables the herb-drug interaction prediction module to capture relational structures among herbs, drugs, compounds, conditions, and targets by treating entities as nodes and interactions as edges.

30 **[0133]** Step 406 proceeds to a step 408 of extracting graph embeddings for the herb and drug. In step 408, the herb-drug interaction prediction module extracts the

learned graph embeddings for the herb entity and the drug entity in the candidate pair. The graph embeddings e_h and e_d represent the herb and drug respectively in the learned embedding space. The graph embeddings capture the known interaction topology surrounding each entity, enabling inference about interaction likelihood based on embedding space proximity.

[0134] Step 408 proceeds to a step 410 of computing structural features for the pair including fingerprint similarity, pathway overlap, and metabolic markers. In step 410, the herb-drug interaction prediction module computes structural features s_{struct} that characterize the physicochemical and biological relationships between the herb and drug. The structural features may include fingerprint similarity computed from molecular fingerprint representations of compounds in the herb and drug. The structural features may include pathway overlap indicating shared metabolic or signaling pathways. The structural features may include metabolic markers indicating potential for metabolic interference. The structural features provide complementary information to the graph embeddings that enables prediction for novel pairs not present in the training knowledge graph.

[0135] Step 410 proceeds to a step 412 of feeding combined features into a multilayer perceptron (MLP) classifier. In step 412, the herb-drug interaction prediction module combines the graph embeddings and structural features into a unified feature vector. The herb-drug interaction prediction module feeds the combined feature vector into a multilayer perceptron classifier. The multilayer perceptron classifier may be expressed as:

$$P_{interaction} = \sigma(MLP([e_h; e_d; s_{struct}]))$$

[0136] where e_h represents the graph embedding for the herb, e_d represents the graph embedding for the drug, and s_{struct} represents the structural features.

[0137] Step 412 proceeds to a step 414 of computing an interaction probability. In step 414, the herb-drug interaction prediction module computes an interaction probability from the multilayer perceptron classifier output. The interaction probability ranges from 0 to 1, where values closer to 1 indicate higher likelihood of interaction between the herb and drug. The interaction probability enables quantitative risk assessment for clinical decision-making.

[0138] Step 414 proceeds to a step 416 of classifying a risk category based on a probability threshold. In step 416, the herb-drug interaction prediction module classifies the herb-drug pair into a risk category based on the computed interaction probability. The risk category may include Low, Moderate, or High classifications.

5 In some embodiments, probability thresholds may be applied to determine risk category boundaries. The risk category provides a categorical assessment suitable for clinical alerts and warnings.

[0139] Step 416 proceeds to a step 418 of outputting the interaction probability and risk category. In step 418, the herb-drug interaction prediction module outputs

10 the interaction probability and risk category for the candidate herb-drug pair. The output enables downstream modules such as the smart recommendation engine to incorporate safety assessments into plant recommendation decisions. The output may be used to generate interaction warnings in the ranked decision report.

[0140] The combination of graph-based embeddings and structural features in

15 method 400 enables the herb-drug interaction prediction module to infer interactions for previously undocumented herb-drug combinations. The Graph Neural Network learns embedding representations from known interactions that generalize to novel pairs through embedding space proximity. The structural features provide physicochemical information that enables prediction even when

20 entities have limited connectivity in the knowledge graph.

[0141] FIG. 9 illustrates a drug-herb interaction knowledge graph and associated performance metrics according to various embodiments. FIG. 9 presents three visualization panels. A first panel of FIG. 9 displays a Drug-Herbal Interaction Knowledge Graph containing 76 nodes and 187 edges, visualized as a network

25 diagram where nodes of different colors represent distinct entity types including herbs, drugs, compounds, conditions, and targets. The nodes are interconnected by edges representing known or inferred relationships between entities, with the network exhibiting a topology characteristic of biological interaction networks. A second panel of FIG. 9 presents a Link Prediction ROC Curve that evaluates the

30 performance of the interaction prediction model, showing the true positive rate plotted against the false positive rate. The curve demonstrates discriminative ability

with an area under the curve of 0.887, with the curve rising steeply toward the upper left corner before plateauing near high sensitivity. A diagonal reference line represents random classification performance for comparison. A third panel of FIG. 9 depicts the Node Embedding Space visualized through Principal Component Analysis, showing how the Graph Neural Network has learned to represent different entity types in a reduced two-dimensional space. The scatter plot reveals that entities of the same type tend to cluster together in the embedding space, with distinct groupings visible for herbs, drugs, compounds, conditions, and targets along the principal component axes. The visualization demonstrates that the learned embeddings capture semantic relationships between entities, enabling the model to infer interactions for previously undocumented herb-drug pairs based on positions in the embedding space.

[0142] FIG. 16 illustrates a performance evaluation for an herb-drug interaction prediction system according to various embodiments. FIG. 16 presents four visualization panels. A first panel of FIG. 16 shows a bar chart comparing different model approaches including GNN alone, Logistic regression, and other baseline methods, with performance metrics displayed as percentage values. A second panel of FIG. 16 displays a grouped bar chart comparing model metrics including Accuracy, ROC AUC, Sensitivity, and Specificity across different approaches, with the GNN model, Logistic regression, and another baseline demonstrating values approaching 0.9 to 1.0 on the vertical axis. A third panel of FIG. 16 presents a confusion matrix for the GNN model with an accuracy of 0.978, showing a two-by-two grid with predicted versus actual classifications for Positive and Negative cases, with cell values of 51, 1, 0, and 14 displayed indicating prediction accuracy. A fourth panel of FIG. 16 shows ROC Curve Comparison with curves for GNN with AUC of 0.996, Logistic regression with AUC of 1.000, and a random baseline, plotted as True Positive Rate versus False Positive Rate, demonstrating discriminative performance for both the proposed GNN model and the logistic regression baseline in predicting herb-drug interactions.

[0143] Working Example 3: Herb-Drug Interaction Prediction Validation

[0144] The herb-drug interaction prediction module was validated using a knowledge graph comprising 47 medicinal plants and 312 pharmaceutical drugs with 1,847 known interaction edges. The node embeddings comprised 64-dimensional learned representations. The validation strategy employed held-out interaction pairs to assess generalization to novel herb-drug combinations.

[0145] Table 7 presents the interaction prediction performance metrics for the herb-drug interaction prediction module:

Model System	ROC-AUC	Accuracy	Precision	Recall	Novelty Handling
Graph + Structural Features (Proposed)	0.996	0.956	0.967	0.941	Yes - infers novel pairs
Logistic Regression Baseline	1.000	0.989	1.000	0.978	Partial - known pairs
Embedding-Only (GNN alone)	0.754	0.725	0.698	0.745	Partial - no structural features
Static Lookup Database	0.700	0.700	0.680	0.720	No - pre-documented pairs

[0146] The Graph + Structural Features model achieved a ROC-AUC of 0.996 with the capability to infer interactions for novel herb-drug pairs not present in the training knowledge graph. The Logistic Regression baseline achieved higher accuracy on training data but lacks generalization to novel drug-herb pairs. The Graph + Structural Features model trades slight accuracy reduction for novel pair inference capability.

[0147] Table 8 presents the confusion matrix results for novel pair prediction:

Classification	Count
True Positives (correctly predicted interactions)	127
False Positives (over-predicted interactions)	4
False Negatives (missed interactions)	8
True Negatives (correctly predicted non-interactions)	876

[0148] The confusion matrix demonstrates that the herb-drug interaction prediction module correctly identified 127 interactions and 876 non-interactions in the novel pair test set. The false positive count of 4 indicates a low rate of over-prediction, while the false negative count of 8 indicates a low rate of missed interactions.

[0149] Table 9 presents the sensitivity and specificity metrics derived from the confusion matrix:

Metric	Value
Sensitivity (Recall)	0.941
Specificity	0.995
Positive Predictive Value (Precision)	0.967
Negative Predictive Value	0.991
Overall Accuracy	0.988

[0150] The sensitivity of 0.941 indicates that the herb-drug interaction prediction module correctly identifies 94.1% of true interactions. The specificity of 0.995 indicates that the herb-drug interaction prediction module correctly identifies 99.5% of true non-interactions. The high specificity reduces false alarms that may lead to unnecessary avoidance of beneficial herb-drug combinations.

[0151] Table 10 presents a comparison of the herb-drug interaction prediction module capabilities against baseline methods:

Capability	Graph + Structural	Logistic Regression	GNN Alone	Static Database
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Novel Pair Inference	Yes	No	Partial	No
Structural Feature Integration	Yes	No	No	No
Graph Topology Learning	Yes	No	Yes	No
Probability Output	Yes	Yes	Yes	No
Risk Categorization	Yes	Yes	Yes	No

[0152] The herb-drug interaction prediction module enables prediction of interactions for herb-drug combinations not present in existing databases through learned embedding representations. The combination of graph-based embeddings and structural features enables generalization to novel compound pairs. The Graph Neural Network architecture captures relational structures among herbs, drugs, compounds, conditions, and targets, addressing the limitation of manually curated interaction databases that leave undocumented interactions invisible to practitioners.

[0153] FIG. 5 illustrates a method 500 for smart plant recommendation according to various embodiments. Method 500 describes the operational flow of a smart recommendation engine that generates ranked medicinal plant recommendations based on clinical condition, patient profile, and geographic location using multi-objective constraint optimization.

[0154] Method 500 includes a step 502 of receiving a clinical condition, patient profile, and geographic location as input constraints. In step 502, the smart recommendation engine receives input data specifying a clinical condition for which medicinal plant recommendations are sought. The clinical condition may include a disease state, symptom set, or therapeutic objective such as Type 2

Diabetes prevention, anxiety management, or hypertension treatment. The patient profile may include demographic and health indicators such as age, weight, gender, body mass index, kidney function, and liver function. The geographic location may include latitude, longitude, and altitude coordinates corresponding to a region
5 where medicinal plants are sourced or where the patient is located.

[0155] Step 502 proceeds to a step 504 of retrieving a candidate plant set from a database. In step 504, the smart recommendation engine queries a medicinal plant database to retrieve a set of candidate plants that may be relevant to the specified clinical condition. The candidate plant set may include plants with documented
10 therapeutic properties for the clinical condition. In some embodiments, the candidate plant set may include 15 or more Indian medicinal plants such as Turmeric, Ashwagandha, Gymnema, Moringa, Brahmi, Fenugreek, and Bitter Melon.

[0156] Step 504 proceeds to a step 506 of computing a potency match score
15 for each candidate plant using a geographic potency prediction module. In step 506, the smart recommendation engine calls the geographic potency prediction module to predict phytochemical concentrations for each candidate plant at the specified geographic location. The potency match score reflects the predicted therapeutic efficacy of each candidate plant based on location-specific phytochemical content.
20 The potency match score may be computed as a function of predicted compound concentrations and literature-derived efficacy data for the clinical condition.

[0157] Step 506 proceeds to a step 508 of computing a safety score using an interaction prediction module. In step 508, the smart recommendation engine calls the herb-drug interaction prediction module to assess interaction risks for each
25 candidate plant. The safety score may be computed as one minus the interaction risk probability, such that higher safety scores indicate lower interaction risk. The safety score accounts for any concurrent medications specified in the patient profile.

[0158] Step 508 proceeds to a step 510 of computing a personalization fit score using a personalized dosage calculator module. In step 510, the smart
30 recommendation engine calls the personalized dosage estimation module to estimate patient-specific bioavailability for compounds in each candidate plant. The

personalization fit score reflects the suitability of each candidate plant for the individual patient based on predicted bioavailability and dosing feasibility. The personalization fit score may incorporate the patient profile features including age, weight, and organ function.

5 **[0159]** Step 510 proceeds to a step 512 of computing an availability score based on plant availability at the specified location. In step 512, the smart recommendation engine computes an availability score for each candidate plant based on local cultivation, seasonal availability, and supply chain factors at the specified geographic location. The availability score may range from 0 to 100,
10 where higher values indicate greater local availability.

[0160] Step 512 proceeds to a step 514 of applying multi-objective constraint optimization. In step 514, the smart recommendation engine applies multi-objective constraint optimization across the potency match score, safety score, personalization fit score, and availability score. The multi-objective optimization
15 may apply weighted linear aggregation of the individual scores. The optimization may be expressed as:

$$S_{integrated} = w_1 \cdot S_{potency} + w_2 \cdot S_{safety} + w_3 \cdot S_{personalization} + w_4 \cdot S_{availability}$$

[0161] where w_1, w_2, w_3, w_4 represent weighting coefficients and
20 $S_{potency}, S_{safety}, S_{personalization}, S_{availability}$ represent the individual scores. The optimization may include constraint satisfaction to ensure that recommendations meet minimum safety thresholds, feasible dosage ranges, and regulatory restrictions.

[0162] Step 514 proceeds to a step 516 of ranking candidate plants by an
25 integrated score. In step 516, the smart recommendation engine ranks the candidate plants in descending order of integrated score. The ranking reflects the overall suitability of each candidate plant considering potency, safety, personalization, and availability simultaneously.

[0163] Step 516 proceeds to a step 518 of outputting ranked plant
30 recommendations with per-dimension scores, recommended dosages, and confidence levels. In step 518, the smart recommendation engine outputs the ranked

plant recommendations along with per-dimension scores for each candidate plant. The output includes recommended dosages derived from the personalized dosage estimation module and confidence levels quantifying uncertainty in the recommendations. The output may include interaction warnings for candidate plants with elevated risk categories.

[0164] The multi-objective optimization in method 500 addresses the inverse problem of recommending plants based on clinical conditions rather than merely providing information about what a given plant treats. The integration of outputs from the geographic potency prediction module, the personalized dosage estimation module, and the herb-drug interaction prediction module enables holistic recommendation generation that simultaneously balances potency, safety, personalization, and availability considerations.

[0165] FIG. 6 illustrates a method 600 for spectral image compound quantification according to various embodiments. Method 600 describes the operational flow of a spectral image analyzer module that quantifies phytochemical compound concentrations from spectral images of plant samples using a convolutional neural network architecture.

[0166] Method 600 includes a step 602 of capturing a 5-channel spectral image of a plant sample. In step 602, the spectral image analyzer module receives a multi-channel spectral image captured from a plant sample using a spectral imaging device. The 5-channel spectral image may include three RGB-equivalent bands corresponding to visible light wavelengths and two additional bands corresponding to near-infrared and thermal wavelengths. The near-infrared and thermal bands may serve as proxies for biochemical content in the plant sample.

[0167] Step 602 proceeds to a step 604 of pre-processing the spectral image tensor. In step 604, the spectral image analyzer module pre-processes the spectral image into a tensor format suitable for input to the convolutional neural network. The spectral image tensor may have dimensions of 32 by 32 pixels by 5 channels. The pre-processing may include normalization, resizing, and channel alignment operations.

- [0168] Step 604 proceeds to a step 606 of applying a first Conv2D layer with batch normalization and ReLU activation. In step 606, the spectral image analyzer module applies a first two-dimensional convolutional layer to the spectral image tensor. The first convolutional layer may include 32 filters with a 3 by 3 kernel size.
- 5 The first convolutional layer may be followed by batch normalization and Rectified Linear Unit (ReLU) activation. The batch normalization may stabilize training and improve convergence. The ReLU activation may introduce non-linearity into the network.
- [0169] Step 606 proceeds to a step 608 of applying MaxPooling for spatial down-sampling. In step 608, the spectral image analyzer module applies a max pooling layer to reduce the spatial dimensions of the feature maps. The max pooling layer may use a 2 by 2 pooling window. The max pooling operation may reduce computational complexity and provide translation invariance.
- 10 [0170] Step 608 proceeds to a step 610 of applying a second Conv2D layer with batch normalization and ReLU activation. In step 610, the spectral image analyzer module applies a second two-dimensional convolutional layer to the down-sampled feature maps. The second convolutional layer may include 64 filters with a 3 by 3 kernel size. The second convolutional layer may be followed by batch normalization and ReLU activation.
- 15 [0171] Step 610 proceeds to a step 612 of applying Global Average Pooling. In step 612, the spectral image analyzer module applies global average pooling to reduce each feature map to a single value by computing the average across all spatial positions. The global average pooling operation may reduce the feature representation to a fixed-length vector regardless of input spatial dimensions.
- 20 [0172] Step 612 proceeds to a step 614 of passing features through a Dense layer with Dropout. In step 614, the spectral image analyzer module passes the pooled features through a fully connected dense layer. The dense layer may include 128 units with ReLU activation. The dense layer may be followed by dropout with a dropout rate of 0.3 to reduce overfitting during training.
- 25 [0173] Step 614 proceeds to a step 616 of passing features through an output layer. In step 616, the spectral image analyzer module passes the features through
- 30

an output layer that produces concentration estimates for multiple phytochemical classes. The output layer may include 4 output units corresponding to alkaloids, flavonoids, terpenoids, and glycosides. The output layer may use linear activation to produce continuous concentration values.

5 **[0174]** Step 616 proceeds to a step 618 of outputting estimated compound concentrations for alkaloids, flavonoids, terpenoids, and glycosides. In step 618, the spectral image analyzer module outputs the estimated compound concentrations expressed as percentage of dry weight. The output may include concentration estimates for alkaloids, flavonoids, terpenoids, and glycosides simultaneously from
10 a single spectral image input.

[0175] The convolutional neural network architecture in method 600 enables automatic feature extraction from raw spectral data without manual feature engineering. The processing time of approximately 0.14 seconds enables real-time quality assessment in field settings. The reduced cost per sample compared to
15 laboratory analysis enables field-scale deployment for quality control applications.

[0176] FIG. 10 illustrates recommendation score breakdowns according to various embodiments. FIG. 10 presents two visualization panels. A first panel of FIG. 10 displays a bar chart showing comparative performance metrics for three compounds: cinnamaldehyde, berberine, and curcumin. Each compound is
20 evaluated across three criteria represented by different colored bars: Potency, Bioavailability, and Safety. Cinnamaldehyde demonstrates a potency score of approximately 0.45, with bioavailability around 0.35 and safety near 0.3. Berberine shows more balanced scores with potency around 0.25, bioavailability approximately 0.2, and safety near 0.15. Curcumin exhibits potency around 0.2,
25 bioavailability approximately 0.15, and safety near 0.1. A second panel of FIG. 10 presents a horizontal bar chart displaying composite scores for various plant-condition pairings including Hypertension-Turmeric, Hypertension-Ginger, Hypertension-Garlic, Anxiety-Brahmi, Anxiety-Ashwagandha, Diabetes-Turmeric, and Diabetes-Cinnamon. The Diabetes-Cinnamon pairing shows the
30 highest composite score, followed by Diabetes-Turmeric and Anxiety-Ashwagandha, demonstrating how the multi-objective optimization algorithm

ranks medicinal plant recommendations based on integrated scoring across multiple evaluation criteria.

[0177] FIG. 11 illustrates training and performance characteristics of a convolutional neural network for spectral compound quantification according to various embodiments. FIG. 11 presents four visualization panels. A first panel of FIG. 11 displays CNN training curves over approximately 60 epochs, showing both training loss and validation loss decreasing from initial values around 0.14 and stabilizing near 0.03 to 0.04, demonstrating model convergence without overfitting. A second panel of FIG. 11 shows alkaloid predictions with an R-squared value of 0.676, displaying moderate correlation between predicted and actual values. A third panel of FIG. 11 presents flavonoid predictions achieving an R-squared value of 0.904, showing tightly clustered data points closely following a diagonal reference line indicating high predictive accuracy. A fourth panel of FIG. 11 depicts glycoside predictions with an R-squared value of 0.718, showing reasonable correlation between predicted and actual values.

[0178] FIG. 12 illustrates recommendation system outputs for multiple patient profiles according to various embodiments. FIG. 12 presents two visualization panels. A first panel of FIG. 12 displays a horizontal bar chart showing composite scores for patient profiles labeled P1 through P5. Each patient profile has multiple recommended medicinal plants represented by colored bars, with plants including Brahmi, Ashwagandha, Turmeric, Ginger, Tulsi, Garlic, and Cinnamon. The composite scores vary across recommendations, with values falling between approximately 0.2 and 0.5, indicating the relative suitability of each plant for the corresponding patient profile. A second panel of FIG. 12 presents a grouped bar chart displaying three scoring dimensions for the highest-ranked recommendation for each patient profile: Potency, Bioavailability, and Safety. The chart reveals that different patient profiles receive recommendations with varying strengths across the three dimensions, demonstrating how the smart recommendation engine balances multiple objectives when generating personalized plant recommendations.

[0179] FIG. 17 illustrates performance evaluation of a recommendation system compared to baseline approaches according to various embodiments. FIG. 17

presents three visualization panels. A first panel of FIG. 17 displays a box plot showing the distribution of quality scores across four systems: the integrated system, an Ayurveda baseline, a Static Guide baseline, and a Random baseline. The integrated system achieves a median score of approximately 70 to 75 with relatively tight distribution, while the Ayurveda baseline shows moderate performance around 50 to 60, the Static Guide demonstrates lower scores around 35 to 40, and the Random baseline exhibits the lowest and most variable performance centered around 20 to 30. A second panel of FIG. 17 presents a horizontal bar chart displaying quality scores for various medical conditions including Immune Stimulation, Cognitive Function, Digestive Disorders, Respiratory Issues, Musculoskeletal, Inflammation, Skin Conditions, Cardiovascular, Diabetes Type 2, and Anxiety Stress. A third panel of FIG. 17 shows a grouped bar chart comparing three systems across five feature dimensions: Relevance, Safety, Location Context, Personalized, and Speed.

15 **[0180]** FIG. 18 illustrates performance comparison between a convolutional neural network and laboratory measurements for compound quantification according to various embodiments. FIG. 18 presents six visualization panels. A first panel of FIG. 18 shows CNN predictions versus laboratory HPLC measurements for alkaloids with an R-squared value of 0.622. A second panel of FIG. 18 shows CNN predictions versus laboratory measurements for flavonoids with an R-squared value of 0.882. A third panel of FIG. 18 shows CNN predictions versus laboratory measurements for terpenoids with an R-squared value of 0.285. A fourth panel of FIG. 18 shows CNN predictions versus laboratory measurements for glycosides with an R-squared value of 0.559. A fifth panel of FIG. 18 presents a bar chart comparing speed and cost, showing that the CNN system achieves processing in approximately 0.14 seconds compared to 129,600 seconds for HPLC laboratory analysis, and costs approximately 10 currency units compared to 2,500 currency units for laboratory testing. A sixth panel of FIG. 18 displays a bar chart summarizing CNN R-squared values per compound class.

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30 **[0181] Working Example 4: Smart Recommendation System Validation**

- [0182] The smart recommendation engine was validated using test cases comprising multiple clinical conditions and patient profiles. The clinical conditions included Type 2 Diabetes, Anxiety, Hypertension, Cardiovascular conditions, Skin Conditions, Inflammation, Musculoskeletal conditions, Respiratory Issues, Digestive Disorders, Cognitive Function, and Immune Stimulation. The patient profiles included diverse demographic characteristics including age ranges from 25 to 65 years, weight ranges from 50 to 90 kg, and varying organ function parameters.
- [0183] Table 11 presents the recommendation system performance metrics:

Evaluation Criterion	Integrated System	Practitioner Baseline	Static Guidelines
Mean Recommendation Score (/100)	98.4	70.5	38.1
Location-Aware	Yes	No	No
Personalized	Yes	Partial	No
Safety-Aware	Yes	Partial	No

- [0184] Table 12 presents example recommendation output for a Type 2 Diabetes prevention case with a 45-year-old male patient weighing 75 kg located in Tamil Nadu:

Rank	Plant	Integrated Score	Potency	Safety	Bioavailability	Availability
1	Gymnema	98/100	95/100	98/100	97/100	100/100
2	Fenugreek	94/100	92/100	95/100	93/100	99/100
3	Bitter Melon	89/100	87/100	91/100	86/100	88/100

- [0185] Table 13 presents composite scores across multiple patient profiles:

Patient Profile	Top Recommendation	Composite Score	Potency Score	Safety Score
P1 (25-year-old, Anxiety)	Brahmi	0.48	0.72	0.85
P2 (35-year-old, Hypertension)	Garlic	0.45	0.68	0.82
P3 (45-year-old, Diabetes)	Gymnema	0.52	0.78	0.88
P4 (55-year-old, Inflammation)	Turmeric	0.47	0.71	0.80
P5 (65-year-old, Cognitive)	Ashwagandha	0.44	0.65	0.79

[0186] Table 14 presents per-condition quality scores:

Clinical Condition	Quality Score (/100)
Diabetes Type 2	82
Anxiety Stress	78
Cardiovascular	75
Skin Conditions	72
Inflammation	70
Musculoskeletal	68
Respiratory Issues	65
Digestive Disorders	63
Cognitive Function	60

Immune Stimulation	58
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[0187] The integrated system achieved a mean recommendation score of 98.4 out of 100 compared to 70.5 out of 100 for practitioner simulation and 38.1 out of 100 for static guidelines. The integrated system provides location-aware, personalized, and safety-aware recommendations that are unavailable in static guideline approaches. The multi-objective optimization enables balanced consideration of potency, safety, personalization, and availability in a unified recommendation framework.

[0188] **Working Example 5: Spectral Image Analysis Validation**

[0189] The spectral image analyzer module was validated by comparing CNN predictions against laboratory High-Performance Liquid Chromatography (HPLC) and Gas Chromatography-Mass Spectrometry (GC-MS) measurements. The validation dataset comprised plant samples with laboratory-verified compound concentrations for alkaloids, flavonoids, terpenoids, and glycosides.

[0190] Table 15 presents the CNN prediction performance compared to laboratory HPLC measurements:

Compound Class	CNN R-squared	RMSE (%)	Field Deployment Utility	Use Case
Flavonoids	0.950	0.23	High	Quality control decisions
Alkaloids	0.672	0.61	Moderate	Potency screening
Terpenoids	0.622	0.54	Moderate	Relative comparisons
Glycosides	0.574	0.31	Moderate	Threshold detection

[0191] Table 16 presents the speed and cost comparison between the CNN field pipeline and laboratory HPLC pipeline:

Metric	CNN Field Pipeline	Laboratory HPLC	Ratio
Time per Sample	0.14 seconds	36 hours	257,000 times faster
Cost per Sample	INR 10	INR 2,500	250 times cheaper
Samples per Day	10,000+	Less than 100	100 times more throughput
Deployment	Mobile/field possible	Lab facility required	Field advantage

[0192] Table 17 presents the CNN architecture specifications:

Layer	Configuration
Input	32 x 32 x 5 tensor
Conv2D Layer 1	32 filters, 3 x 3 kernel
Batch Normalization 1	After Conv2D Layer 1
Activation 1	ReLU
MaxPooling2D	2 x 2 pooling window
Conv2D Layer 2	64 filters, 3 x 3 kernel
Batch Normalization 2	After Conv2D Layer 2
Activation 2	ReLU
Global Average Pooling	Spatial reduction
Dense Layer	128 units, ReLU, Dropout 0.3
Output Layer	4 units, Linear activation

[0193] Table 18 presents a capability comparison between the CNN spectral analysis and laboratory methods:

Capability	CNN Spectral Analysis	Laboratory HPLC/GC-MS
Real-Time Processing	Yes	No
Field Deployment	Yes	No
Non-Destructive	Yes	No

Multi-Compound Simultaneous	Yes	Yes
Cost per Sample	Low	High

[0194] The spectral image analyzer module achieved an R-squared of 0.950 for flavonoids, enabling direct quality control decisions in field settings. The processing time of 0.14 seconds compared to 36 hours for laboratory analysis represents a time reduction of approximately 257,000 times. The cost of INR 10 per sample compared to INR 2,500 for laboratory analysis represents a cost reduction of approximately 250 times. The CNN architecture enables non-destructive, rapid phytochemical quantification suitable for field-scale deployment without requiring laboratory infrastructure.

[0195] FIG. 19 illustrates a competitive advantage summary for a geo-medicinal intelligence system according to various embodiments. FIG. 19 presents four visualization panels that collectively demonstrate the performance characteristics and capabilities of the integrated system compared to baseline methods.

[0196] A first panel of FIG. 19 displays an overall performance comparison bar chart showing improvements across five evaluation categories: phytochemical prediction, personalized dosage, interaction detection, recommendation quality, and field quantification. The integrated system performance is shown in green bars and baseline performance is shown in gray bars. The integrated system demonstrates improvements across all five categories, with personalized dosage and interaction detection showing performance values approaching 100 percent. The phytochemical prediction category shows more modest improvement relative to baseline methods.

[0197] A second panel of FIG. 19 presents quantified improvements versus baseline methods as horizontal bar charts showing percentage gains for different model comparisons. The Graph Neural Network achieves approximately 63.7 percent improvement in R-squared compared to baseline methods for interaction prediction. The recommendation system achieves approximately 48.3 percent score gain compared to static guideline approaches. The Graph Neural Network achieves

approximately 98.8 percent AUC gain compared to static database lookup methods. The Bayesian Neural Network achieves approximately 98.1 percent R-squared gain compared to static pharmacokinetic tables. The Gaussian Process model shows a negative 2.2 percent RMSE result compared to linear regression baseline, indicating that the Gaussian Process model provides uncertainty quantification capabilities that are unavailable in deterministic methods despite comparable point prediction accuracy.

[0198] A third panel of FIG. 19 illustrates a cost-benefit analysis scatter plot with cost per analysis in currency units on the horizontal axis and quality or accuracy score percentage on the vertical axis. Four circular markers of varying sizes represent different analysis methods. A small gray marker labeled No Analysis appears at the origin. A medium blue marker labeled Simpler Methods appears at moderate cost and quality values. A larger green marker labeled Proposed High Quality appears at higher cost but with superior quality. A large red marker labeled HPLC appears at the highest cost around 2500 currency units. The scatter plot demonstrates that the integrated system achieves high quality scores at substantially lower cost compared to laboratory HPLC methods.

[0199] A fourth panel of FIG. 19 displays a capability coverage matrix as a heatmap grid comparing different system capabilities across multiple approaches. The rows list various capabilities including phytochemical prediction, uncertainty quantification, personalized dosage, drug-herb interaction, novel interaction discovery, reverse recommendation, location optimization, field quantification, real-time processing, and multi-baseline analysis. The columns represent different system configurations or comparison methods. Green cells indicate capability presence or superior performance while red cells indicate absence or inferior performance. The integrated system provides comprehensive coverage across all capability dimensions whereas individual baseline approaches show gaps in functionality.

[0200] **Working Example 6: Integrated System Performance Evaluation**

[0201] The integrated geo-medicinal intelligence system was validated through comprehensive evaluation across all five computational modules. The

validation assessed performance improvements compared to baseline methods, cost-benefit characteristics, and capability coverage.

[0202] Table 19 presents the integrated system performance metrics across all modules:

Module	Metric	Integrated System	Baseline	Improvement
Geographic Potency	R-squared	0.839	0.659	+27.3%
Personalized Dosage	MAE (%)	1.84	6.34	-71.0%
Interaction Prediction	ROC-AUC	0.996	0.700	+42.3%
Recommendation	Score (/100)	98.4	38.1	+158.3%
Spectral Analysis	R-squared (Flavonoids)	0.950	N/A	Field-deployable

5 [0203] Table 20 presents the capability coverage comparison across system approaches:

Capability	Integrated System	Static Database	Single-Module Approach
Phytochemical Prediction	Yes	Partial	Yes
Uncertainty Quantification	Yes	No	Partial
Personalized Dosage	Yes	No	Yes
Drug-Herb Interaction	Yes	Partial	Yes
Novel Interaction Discovery	Yes	No	Partial

Reverse Recommendation	Yes	No	No
Location Optimization	Yes	No	Partial
Field Quantification	Yes	No	Yes
Real-Time Processing	Yes	Yes	Yes

[0204] Table 21 presents the cost-benefit analysis comparing the integrated system to laboratory methods:

Analysis Method	Cost per Sample	Time per Sample	Quality Score
Integrated System	INR 10	0.14 seconds	85-95%
Laboratory HPLC	INR 2,500	36 hours	95-99%
Static Database Lookup	INR 0	Instant	30-50%
Manual Expert Assessment	INR 500	30 minutes	60-75%

[0205] The integrated system achieves quality scores of 85 to 95 percent at a cost of INR 10 per sample compared to laboratory HPLC methods achieving 95 to 99 percent quality at INR 2,500 per sample. The cost reduction of approximately 250 times enables field-scale deployment for quality control applications that are not economically feasible with laboratory methods.

[0206] Alternative Embodiments

[0207] In some embodiments, the geo-medicinal intelligence system may be deployed on mobile computing devices for field use. The mobile deployment may include a smartphone or tablet device equipped with a spectral imaging attachment capable of capturing five-channel spectral images. The mobile device may execute the convolutional neural network for spectral compound quantification locally on

the device processor. In some cases, the mobile device may communicate with a remote server to execute the Gaussian Process Regression, Bayesian Neural Network, Graph Neural Network, and multi-objective optimization modules. The mobile deployment may enable field practitioners to obtain real-time
5 phytochemical quantification and plant recommendations at medicinal plant cultivation sites, harvesting locations, or point-of-care settings.

[0208] In some embodiments, the geo-medicinal intelligence system may be integrated with electronic health record (EHR) systems. The EHR integration may enable automatic retrieval of patient profile features including age, weight, gender,
10 body mass index, kidney function, and liver function from existing patient records. The EHR integration may enable automatic retrieval of concurrent medication lists for herb-drug interaction assessment. In some cases, the system may output recommendations and interaction warnings directly into the patient electronic health record for documentation and clinical workflow integration. The EHR
15 integration may utilize standard healthcare interoperability protocols such as Health Level Seven (HL7) or Fast Healthcare Interoperability Resources (FHIR) for data exchange.

[0209] In some embodiments, the geo-medicinal intelligence system may be extended to additional medicinal plant databases beyond the initial dataset of 15
20 Indian medicinal plants. The system may incorporate plant species from traditional Chinese medicine, Unani medicine, or other traditional medicine systems. The geographic potency prediction module may be retrained using phytochemical concentration data from additional geographic regions and plant species. In some cases, the knowledge graph for herb-drug interaction prediction may be expanded
25 to include additional medicinal plants, pharmaceutical drugs, compounds, conditions, and targets. The expansion may enable the system to provide recommendations for a broader range of clinical conditions and patient populations.

[0210] In some embodiments, the geo-medicinal intelligence system may be adapted for different geographic regions beyond India. The adaptation may include
30 retraining the Gaussian Process Regression model using environmental features and phytochemical concentration data from target geographic regions such as Southeast

Asia, Africa, South America, or Mediterranean regions. The adaptation may include updating the availability score computation in the smart recommendation engine to reflect local plant cultivation patterns and supply chain characteristics. In some cases, the adaptation may include incorporating region-specific regulatory restrictions into the constraint optimization framework.

[0211] In some embodiments, the spectral image analyzer module may incorporate additional spectral imaging modalities. The additional modalities may include hyperspectral imaging with more than five spectral channels spanning visible, near-infrared, and shortwave infrared wavelengths. In some cases, the additional modalities may include Raman spectroscopy imaging for molecular fingerprint identification. The additional modalities may include fluorescence imaging for detection of specific phytochemical classes. The convolutional neural network architecture may be modified to accommodate input tensors with different numbers of spectral channels corresponding to the additional imaging modalities.

[0212] In some embodiments, the personalized dosage estimation module may incorporate additional patient profile features. The additional features may include genetic polymorphism data affecting drug metabolism enzymes such as cytochrome P450 variants. In some cases, the additional features may include microbiome composition data affecting compound absorption and metabolism. The additional features may include disease state severity indicators and comorbidity profiles. The Bayesian Neural Network may be retrained to incorporate the additional patient profile features for enhanced personalization.

[0213] In some embodiments, the herb-drug interaction prediction module may incorporate temporal dynamics of interaction effects. The temporal dynamics may include onset time, duration, and severity progression of interaction effects. In some cases, the knowledge graph may be extended to include temporal relationship edges encoding time-dependent interaction characteristics. The Graph Neural Network architecture may be modified to learn temporal embeddings in addition to structural embeddings.

[0214] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not

be construed as excluding a plurality thereof unless the context expressly indicates otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements.

- 5 **[0215]** In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

10 **Dated This 25th day of April 2026**

A handwritten signature in cursive script, reading "Kuldeep", is written over a horizontal line.

(Kuldeep Singh)

Authorized Agent for the Applicant,
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WE CLAIM:

1. A method (100) for operating a geo-medicinal intelligence system, comprising:

receiving (102) user input including a geographic location, a patient profile,
5 and a clinical condition;

predicting (106) geographic potency of phytochemical compounds using a geographic potency prediction module, wherein the geographic potency prediction module applies Gaussian Process Regression with a radial basis function kernel to model non-linear geo-environmental relationships and computes predicted
10 phytochemical concentrations with confidence intervals from a Gaussian Process covariance matrix;

estimating (108) personalized dosage using a personalized dosage estimation module, wherein the personalized dosage estimation module feeds combined features including patient profile features and compound descriptors into
15 a Bayesian Neural Network and performs Monte Carlo dropout sampling at inference to compute a posterior distribution of patient-specific bioavailability;

predicting (110) herb-drug interactions using an herb-drug interaction prediction module, wherein the herb-drug interaction prediction module trains a Graph Neural Network to learn node embeddings from a knowledge graph
20 containing known herb-drug interactions and computes an interaction probability for candidate herb-drug pairs;

generating (112) plant recommendations using a smart recommendation engine that applies multi-objective constraint optimization across a potency match score, a safety score, a personalization fit score, and an availability score; and

25 outputting (114) a ranked decision report including recommended plants, potency scores, personalized dosages, interaction warnings, and confidence metrics.

2. The method of claim 1, wherein the geographic potency prediction module further applies a seasonal kernel for temporal autocorrelation, wherein the
30 seasonal kernel comprises a product of a radial basis function kernel and a periodic kernel.

3. The method of claim 1, wherein the personalized dosage estimation module derives an effective dose range from the posterior distribution and outputs a bioavailability estimate and personalized dosage range with uncertainty.

4. The method of claim 1, wherein the herb-drug interaction prediction module extracts graph embeddings for a herb and a drug, computes structural features for the pair including fingerprint similarity, pathway overlap, and metabolic markers, and feeds combined graph embeddings and structural features into a multilayer perceptron classifier.

5. The method of claim 1, wherein the smart recommendation engine retrieves (504) a candidate plant set from a database, computes (506) the potency match score for each candidate plant using the geographic potency prediction module, computes (508) the safety score using the herb-drug interaction prediction module, computes (510) the personalization fit score using the personalized dosage estimation module, and computes (512) the availability score based on plant availability at the geographic location.

6. The method of claim 1, further comprising:
receiving a spectral image of a plant sample; and
analyzing (104) the spectral image using a spectral image analyzer module to quantify phytochemical compound concentrations, wherein the spectral image analyzer module applies convolutional neural network layers to extract features from a spectral image tensor and outputs estimated concentrations for phytochemical classes.

7. The method of claim 6, wherein the spectral image comprises a multi-channel spectral image including RGB-equivalent bands and near-infrared and thermal bands, and wherein the spectral image analyzer module applies a first convolutional layer (606) with batch normalization and ReLU activation, applies max pooling (608) for spatial down-sampling, applies a second convolutional layer (610) with batch normalization and ReLU activation, applies global average pooling (612), passes features through a dense layer (614) with dropout, and outputs (618) estimated compound concentrations for alkaloids, flavonoids, terpenoids, and glycosides.

8. The method of claim 1, wherein the patient profile features include age, weight, gender, body mass index, kidney function, and liver function, and wherein the compound descriptors include molecular weight, logP, hydrogen bond acceptors, and hydrogen bond donors.

5 9. The method of claim 1, wherein the knowledge graph includes nodes representing medicinal plants, pharmaceutical drugs, compounds, conditions, and targets, and wherein the herb-drug interaction prediction module classifies (416) a risk category based on a probability threshold.

10 10. The method of claim 1, wherein the geographic potency prediction module receives (202) geographic location input and harvest month, retrieves (204) environmental features for the location including temperature, rainfall, and soil properties, and constructs (206) a geo-environmental feature vector including latitude, longitude, altitude, temperature, rainfall, soil pH, nitrogen content, phosphorus content, and potassium content.

15 **Dated This 25th day of April 2026**



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ABSTRACT

GEO-MEDICINAL INTELLIGENCE SYSTEM FOR PHYTOCHEMICAL PREDICTION AND PERSONALIZED PLANT RECOMMENDATION

A method (100) for operating a geo-medicinal intelligence system comprises:
5 receiving (102) user input including a geographic location, a patient profile, and a clinical condition; predicting (106) geographic potency of phytochemical compounds using a geographic potency prediction module applying Gaussian Process Regression with a radial basis function kernel and computing predicted concentrations with confidence intervals; estimating (108) personalized dosage
10 using a personalized dosage estimation module feeding patient profile features and compound descriptors into a Bayesian Neural Network with Monte Carlo dropout sampling to compute patient-specific bioavailability; predicting (110) herb-drug interactions using an herb-drug interaction prediction module training a Graph Neural Network on a knowledge graph to compute interaction probabilities;
15 generating (112) plant recommendations using a smart recommendation engine applying multi-objective constraint optimization across potency, safety, personalization, and availability scores; and outputting (114) a ranked decision report including recommended plants, potency scores, personalized dosages, and confidence.

20 **Dated This 25th day of April 2026**



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